A-A-51127
29 June 1984

COMMERCIAL ITEM DESCRIPTION

HOLDER, KNURLING TOOL, LATHE

The General Services Administration has authorized the use of this commercial item description in preference to Type V tool holder of Federal Specification OO-H-581.

This description covers tool holders that are designed to hold and support knurling tools that are used on bench and engine lathes.

Salient characteristics.

Types. The following types of tool holders are covered.

Type I - Holder, knurling tool, single, lathe (Table I)
Type II - Holder, knurling tool, multiple, lathe (Table II)

Dimensions. The dimensions of tool holders and knurls size accommodated shall conform to tables I and II. Knurl styles shall conform to table III. All tolerances of dimensions shall conform to the tolerances as specified on the manufacturers part drawing.

TABLE I. Type I - Single knurling tool holder.

Tool holder number	Thickness of shank; inches	Height of shank; inches	Length of shank; inches, min.	Knurl dimensions		
				Outside diameter inches	Thickness of knurl; inches	Hole diameter inches
E-S-4	5/16	3/4	2-15/16	5/8	5/16	7/32
E-S-10	3/8	7/8	3-1/4	5/8	5/16	7/32
E-S-11	1/2	1-1/8	3-3/4	3/4	3/8	1/4
E-S-12	5/8	1-3/8	5	3/4	3/8	1/4
E-S-13	7/8	1-3/4	5-3/4	1	3/8	5/16

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TABLE II. Type II - multiple knurling tool holder.

Tool holder number	Thickness of shank; inches	Height of shank; inches	Length of shank; inches, min.	Knurl dimensions		
				Outside diameter; inches	Thickness of knurl; inches	Hole diameter; inches
F-R-20	3/8	7/8	2-1/2	5/8	5/16	7/32
F-R-21	1/2	1-1/8	3-1/8	3/4	3/8	1/4
F-R-22	5/8	1-3/8	3-1/8	3/4	3/8	1/4

TABLE III. Knurl styles available.

Knurl style	Teeth per inch	Bevel degree/ full face	Knurl pattern
K-14-A	14	45°	diamond
K-14-B	14	45°	straight line
K-14-C	14	full face	diamond
K-14-D	14	full face	straight line
K-21-A	21	45°	diamond
K-21-B	21	45°	straight line
K-21-C	21	full face	diamond
K-21-D	21	full face	straight line
K-33-A	33	45°	diamond
K-33-B	33	45°	straight line
K-33-C	33	full face	diamond
K-33-D	33	full face	straight line

Material and hardness. The holder shall be of drop-forged steel, heat-treated to a Rockwell "C" 40 to 48 hardness or case-hardened steel to a Rockwell "C" 50 to 55 hardness, resistant to wearing action of knurls and have means for the mounting and replacement of the knurls.

Support pins. The support pins shall be of steel, heat-treated and finished to properly support and permit free rotation of knurls or rotating heads.

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Knurls. The knurls shall be of tool steel hardened to a Rockwell "C" 63 to 67. Knurls shall be furnished in pairs in either diamond or straight line pattern, either coarse (14 teeth per inch), medium (21 teeth per inch), or fine (33 teeth per inch), either 45 degree bevel to depth of teeth or full face conforming to ANSI B94.6 latest edition with the styles in accordance with table III this description. One pair of style K-21-A knurls of the proper size in accordance with table I will be furnished with each type I tool holder unless otherwise specified in the contract. One pair of each style K-14-A, K-21-A and K-33-A of the proper size in accordance with table II shall be furnished with each type II tool holder unless otherwise specified in the contract.

Measurement system. All dimensions in this commercial item description are given in the English system of measurement. These dimensions can be converted to Metric system equivalents by the use of ANSI/IEEE 268 for Department of Defense use or FED-STD-376 for Civil Agency use.

Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in this description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Acceptance inspection. The supplier shall randomly select the number of samples specified in table IV that is applicable to the quantity of items being purchased. Each sample shall be visually inspected, measured, and tested, by the supplier, to determine conformance with all requirements specified herein. Any sample failing to meet all requirements specified herein shall be considered a defective item. If the number of defective items equals, or is less than the quantity specified under "Acceptance, number defective" column of table IV, the supplier shall replace the defective samples with items that meet all of the requirements specified herein and the entire lot shall be considered acceptable for shipment. If the number of defective samples equals, or exceeds, the quantity specified in the "Rejection, number defective" column of table IV, the entire lot shall be considered defective and unsuitable for acceptance.

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TABLE IV . Sampling plan.

Quantity of items being purchased	Sample size	Acceptance, number defective	Rejection, number defective
2-8	2	0	1
9-15	5	0	1
16-25	8	0	1
26-50	13	1	2
51-90	20	1	2
91-150	32	2	3
151-280	50	3	4
281-500	80	5	6

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling and marking. Unless otherwise specified in the contract or order, preservation, packaging, packing, labeling and marking shall be in accordance with ASTM D-3951. When specified in the contract or order, preservation, packaging, packing, labeling and marking shall be in accordance with Federal Specification PPP-T-1150, level as specified.

Notes. Purchaser should specify quantity, type and tool holder number and quantity and knurl style with the size of knurls conforming to tables I and II for the tool holder number required.

ANSI B94.6 and ANSI/IEEE 268 are available from American National Standards Institute, 1430 Broadway, New York, NY 10018.

ASTM-D-3951 is available from American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

FED-STD-376 and Federal Specification PPP-T-1150 should be obtained from the contracting activity or as directed by the contracting officer.

Type I tool holder, this description, replaces type V, class 1 of Federal Specification 00-H-581 and type II, this description, replaces type V, class 2 of Federal Specification 00-H-581.

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MILITARY INTERESTS:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Air Force - 84
DLA - GS

User Activity

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY

DLA - IP

Project Number 3460-0114

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